

Aoyu Gong

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Information	🌐 Website 📄 Scholar 📖 ResearchGate 🐙 GitHub 🔗 LinkedIn	✉ aoyu.gong@epfl.ch
Educational Background	Doctor of Philosophy , Networking Systems <i>Sep. 2024 – Present</i> 🏛 École Polytechnique Fédérale de Lausanne , Lausanne, Switzerland • Advisor: Haitham Hassanieh Master of Science , Communication Systems <i>Sep. 2021 – July 2024</i> 🏛 École Polytechnique Fédérale de Lausanne , Lausanne, Switzerland • GPA: 5.76/6.00 Bachelor of Engineering , Communication Engineering <i>Sep. 2015 – June 2019</i> 🏛 Nanjing University of Science and Technology , Nanjing, China • GPA: 3.90/4.00	
Work Experience	Research Intern , Network and Cloud Systems Research Group <i>Aug. 2023 – Feb. 2024</i> 🏢 Max Planck Institute for Informatics , Saarbrücken, Germany • Topic: Understand short-form (TikTok/YouTube) video content using multimodal models Research Assistant , Visual Intelligence for Transportation Laboratory <i>Mar. 2023 – May 2023</i> 🏢 École Polytechnique Fédérale de Lausanne , Lausanne, Switzerland • Topic: Rethink traditional operations research problems with deep reinforcement learning	
Selected Publications	* stands for co-primary first authors. A full list can be found here .	
	13. SIGCOMM '26 Aoyu Gong* , Raphael Cannatà*, Arman Maghsoudnia, Néstor Lomba Lomba, Dan Mihai Dumitriu, Haitham Hassanieh, “ SAGE: A Real-Time AI System for Reducing Latency in NextG Cellular Networks ,” <i>Proceedings of the ACM SIGCOMM 2026 Conference</i> , 2026	
	12. OpenRIT6G '25 Aoyu Gong , Arman Maghsoudnia, Raphael Cannatà, Eduard Vlad, Néstor Lomba Lomba, Dan Mihai Dumitriu, Haitham Hassanieh, “ Towards URLLC with Open-Source 5G Software ,” <i>Proceedings of the 1st Workshop on Open Research Infrastructures and Toolkits for 6G</i> , 2025	
	11. TCOM Yuqing Zhu, Yiwen Zhu, Aoyu Gong , Yan Lin, Yuan-Hsuan Lo, Yijin Zhang, “ Age-Gain-Dependent Random Access for Event-Driven Periodic Updating ,” <i>IEEE Transactions on Communications</i> , vol. 73, no. 8, pp. 5774–5787, 2025	
	10. ICWSM '25 Aoyu Gong , Sepehr Mousavi, Yiting Xia, Savvas Zannettou, “ ClipMind: A Framework for Auditing Short-Format Video Recommendations Using Multimodal AI Models ,” in <i>Proceedings of the 19th International AAAI Conference on Web and Social Media</i> , 2025	
	9. MobiCom '24 Raphael Cannatà, Aoyu Gong , Arman Maghsoudnia, Dan Mihai Dumitriu, Haitham Hassanieh, “ SliceGuard: Secure and Dynamic 5G RAN Slicing with WebAssembly ,” in <i>Proceedings of the 30th Annual International Conference on Mobile Computing and</i>	

8. HotNets '24 Arman Maghsoudnia, Eduard Vlad, **Aoyu Gong**, Dan Mihai Dumitriu, Haitham Hassanieh, “[Ultra-Reliable Low-Latency in 5G: A Close Reality or a Distant Goal?](#),” in *Proceedings of the 23rd ACM Workshop on Hot Topics in Networks*, 2024
7. SIGCOMM '24 Jialong Li, Haotian Gong, Federico De Marchi, **Aoyu Gong**, Yiming Lei, Wei Bai, Yiting Xia, “[Uniform-Cost Multi-Path Routing for Reconfigurable Data Center Networks](#),” in *Proceedings of the ACM SIGCOMM 2024 Conference*, 2024
6. TVT **Aoyu Gong**, Yuan-Hsuan Lo, Yan Lin, Yijin Zhang, “[Dynamic Random Access Without Observation Under Deadline-Constrained Periodic Traffic](#),” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 1, pp. 1503–1508, 2024
5. TCOM Yijin Zhang*, **Aoyu Gong***, Lei Deng, Yuan-Hsun Lo, Yan Lin, Jun Li, “[Achieving Maximum Urgency-Dependent Throughput in Random Access](#),” *IEEE Transactions on Communications*, vol. 71, no. 11, pp. 6435–6450, 2023
4. TCOM Jingwei Liu, Rui Zhang, **Aoyu Gong**, He Chen, “[Optimizing Age of Information in Wireless Uplink Networks with Partial Observations](#),” *IEEE Transactions on Communications*, vol. 71, no. 7, pp. 4105–4118, 2023
3. TNSE **Aoyu Gong**, Yijin Zhang, Lei Deng, Fang Liu, Jun Li, Feng Shu, “[Dynamic Optimization of Random Access in Deadline-Constrained Broadcasting](#),” *IEEE Transactions on Network Science and Engineering*, vol. 10, no. 4, pp. 2059–2073, 2023
2. GLOBECOM '20 **Aoyu Gong***, Tong Zhang*, He Chen, Yijin Zhang. “[Age-of-Information-Based Scheduling in Multiuser Uplinks with Stochastic Arrivals: A POMDP Approach](#),” in *Proceedings of the 2020 IEEE Global Communications Conference*, 2020
1. TCOM Yijin Zhang, **Aoyu Gong**, Yuan-Hsun Lo, Jun Li, Feng Shu, Wing Shing Wong. “[Generalized \$p\$ -Persistent CSMA for Asynchronous Multiple-Packet Reception](#),” *IEEE Transactions on Communications*, vol. 67, no. 10, pp. 6966–6979, 2019

Awards & Honors

Best Demo Award	Nov. 2024
• Awarded by ACM MobiCom '24	
EDIC Fellowship	Feb. 2024
• Awarded by École Polytechnique Fédérale de Lausanne	
Provincial Best Bachelor Thesis Award	Sep. 2020
• Awarded by Jiangsu Provincial Department of Education (across all disciplines)	
Bachelor of Engineering with Honors	Jun. 2019
• Awarded by Nanjing University of Science and Technology (top 5%)	
NJUST Best Bachelor Thesis Award	Jun. 2019
• Awarded by Nanjing University of Science and Technology (across all disciplines)	
National Scholarship (Awarded Three Times)	Nov. 2018 / Nov. 2017 / Nov. 2016
• Awarded by Ministry of Education of the People's Republic of China (top 1%)	
NJUST Special-Class Scholarship (Awarded Six Times)	Mar. & Sep. 2018 / 2017 / 2016
• Awarded by Nanjing University of Science and Technology (top 1%)	

Interests & Skills

Area of Interest:

- Networked Systems, Wireless Communication, Machine Learning, Stochastic Control

Computer Skills:

- C/C++, Python, MATLAB, VHDL